

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The area of a rectangular banner is **2,661** square inches. The banner's length x , in inches, is **24** inches longer than its width, in inches. Which equation represents this situation?

Answer

- A. $0 = x^2 - 24x - 2,661$
- B. $0 = x^2 - 24x + 2,661$
- C. $0 = x^2 + 24x - 2,661$
- D. $0 = x^2 + 24x + 2,661$

Correct Answer: A

Rationale

Choice A is correct. It’s given that the banner’s length x , in inches, is **24** inches longer than its width, in inches. It follows that the banner’s width, in inches, can be represented by the expression $x - 24$. The area of a rectangle is the product of its length and its width. It's given that the area of the banner is **2,661** square inches, so it follows that $2,661 = x(x - 24)$, or $2,661 = x^2 - 24x$. Subtracting **2,661** from each side of this equation yields $0 = x^2 - 24x - 2,661$. Therefore, the equation that represents this situation is $0 = x^2 - 24x - 2,661$.

Choice B is incorrect and may result from representing the width, in inches, of the banner as $24 - x$, rather than $x - 24$.

Choice C is incorrect and may result from representing the width, in inches, of the banner as $x + 24$, rather than $x - 24$.

Choice D is incorrect and may result from conceptual or calculation errors.